



RESEARCH ARTICLE

Image Stitching using AKAZE Features

Surendra Kumar Sharma^{1,2}  · Kamal Jain²

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Abstract

Accelerated KAZE (AKAZE) is a multi-scale 2D feature detection and description algorithm in nonlinear scale spaces proposed recently. This paper presents an image stitching algorithm which uses a feature detection and description algorithm; AKAZE and an image blending algorithm; weighted average blending. The whole process is divided into the following steps: First of all, detect feature points in the image and then get feature descriptors of detected points using AKAZE. Next, obtain corresponding matching pairs by using K-NN (K nearest neighbors) algorithm and remove the false matched points by MSAC (M-estimator SAmple Consensus) algorithm. MSAC is a variant of the RANSAC (Random Sample Consensus) algorithm and more accurate than RANSAC. Thereafter, calculate the homography matrix from correct matches. At last, blend the images by using weighted average blending algorithm. Comparison of proposed AKAZE-based algorithm with SIFT-, SURF- and ORB-based algorithms is also presented. According to the experiments, the proposed AKAZE-based image stitching algorithm minimizes stitching seam and generates a perfect stitched image, and also this algorithm is faster than previous algorithms.

Keywords Image stitching · AKAZE · Image quality · Panoramic image

Introduction

Panoramic images or panoramas are wide-angle images that can produce a horizontal field of view up to 360°. Panoramas can be obtained by stitching multiple images together using image stitching algorithm. It is a well-known method of constructively extending field view of the camera. Panoramic imaging softwares implement image stitching algorithm to create large panoramas. Several panorama softwares have been evolved over the last decade. A comparison of commonly used panoramic softwares is provided in (Sharma et al. 2019). Image stitching algorithms have been used in many applications: aerial and satellite image mosaics (Wang et al. 2014; Truong et al. 2018; Tsai and Huang 2004), SAR image mosaicking (Benoit and Thierry 1995; Kwok et al. 1990) video frame

mosaicking (Majumdar et al. 2002), 3D image stitching, virtual tourism, augmented reality (Masoud et al. 2016), and real state are few of them.

Generally, image stitching algorithms consists of two main steps: image alignment and image blending. The development of image stitching technology typically depends on the innovations of these two aspects. Image alignment is used to obtain the motion relationship by detecting and matching feature points across two images or multiple images. It directly associates with the speed and success rate of image stitching process. In recent years, different feature detection–description algorithms are emerged. However, most of them are not suitable for stitching the images because of their speed and a greater number of false matches which degrades the quality of stitched image.

Thus, there is a requirement of a detection–description algorithm which can ensure high quality in image stitching with better computational latency. This paper uses AKAZE algorithm, which is one of the feature detection–description algorithm with high quality of detected points. Some commonly used feature detection–description algorithms are SIFT, SURF and ORB. Many studies are conducted to

✉ Surendra Kumar Sharma
surendra123sharma@gmail.com

¹ Urban and Regional Studies Department, Indian Institute of Remote Sensing Dehradun, Dehradun, Uttarakhand, India

² Department of Civil Engineering, Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India

compare the quality of these popular feature detection algorithms.

The aim of this study is to propose a unique image stitching algorithm which combines AKAZE and weighted average blending algorithms. In the first objective of the paper, AKAZE algorithm was used to detect and extract feature points in a pair of images, and MSAC algorithm was used to obtain the correct matched points from extracted feature points. Weighted average blending was applied to eliminate the stitch seam and illumination variations. In the second objective, proposed AKAZE-based algorithm is compared with SIFT-, SURF- and ORB-based stitching algorithms. A number of datasets consisting of different types of scenes are used for the evaluation. Thus, the research presented in the current paper contributes toward the development of more optimized image stitching algorithms which can ensure the higher quality and better computational time.

This paper is organized in the following way: “[Related Work](#)” section presents the underlying knowledge of AKAZE and related work which uses other feature detection–description algorithms for image stitching process. “[Material and Method](#)” section describes dataset, methodology and working environment. “[Results and Discussion](#)” section presents the experimental results and discussions. Finally, “[Conclusions](#)” section concludes the paper.

Related Work

Image stitching has been well studied in the fields of computer vision and graphics. Image stitching algorithms can be categorized into two broad horizons. The first is the direct method (Irani and Anandan 2000; Szeliski et al. 1980), and the second one is feature based (Zoghلامي et al. 1997; Mclauchlan and Jaenicke 2002). A good survey on image stitching can be found in (Szeliski 2006).

The feature-based techniques primarily consist of four steps: feature detection, feature matching, image warping (transformation) and image blending. In 2004, David G. Lowe proposed a feature detection and extraction algorithm known as SIFT (Scale Invariant Feature Transform) (Lowe 2004). SIFT is used in many computer vision applications. In 2007, Brown and Lowe proposed automatic panorama image stitching algorithm based on SIFT features (Brown and Lowe 2007). A vast amount of literature is available which uses SIFT for image stitching. SIFT-based image stitching algorithms are particularly suitable for stitching high-resolution images under variety of changes (rotation, scale, affine, etc.); however, it has the problem of image mismatch and computationally

expensive. A comparative study of SIFT and its variants is presented in (Wu et al. 2013).

Bay et al. (2008) proposed Speeded-Up Robust Features (SURF) algorithm based on SIFT. This method improves the speed of feature extraction. However, it is not accurate in terms of the stitched output of the algorithm and it may produce large deviation and ghosting effects for stitching multiple images (Lin et al. 2015). Bind et al. (2013) used both SIFT and SURF algorithms for image stitching to take advantage of both the methods, but this takes more time. A comprehensive review of image stitching algorithms based on different feature detection and extraction methods is provided by Ghosh and Kaabouch (2016).

Rublee et al. (2011) presented ORB (Oriented FAST and Rotated BRIEF) an efficient alternative to SIFT and SURF. ORB is a combination of FAST detector (Rosten and Drummond 2006) and BRIEF descriptor (Calonder et al. 2010) with various refinements to increase the performance. A few literatures which use ORB for image stitching applications are available in (Elmogly and Elbakry 2014, 2015; Jeon et al. 2016; Wang et al. 2017). ORB reduces the execution time but at the cost of accuracy. Mehmet (2015) presented a performance evaluation of different feature extraction algorithms on street images. Ali and Salman (2018) compared some of the popular feature detectors and descriptors based on execution time. Andersson and Marquez compared SIFT, AKAZE and ORB for object detection application (2016). Jolhip et al. (2017) compared SIFT and SURF for aerial image stitching.

AKAZE (Accelerated-KAZE)

Alcantarilla et al. (2013) presented a fast and multi-scale feature detection and description algorithm named AKAZE (Accelerated-KAZE). It is an accelerated variation of the KAZE algorithm, presented by the authors in 2012 (Alcantarilla et al. 2012). Earlier proposed approaches which detect and describe features in nonlinear scale space are taking more time as creating the nonlinear scale space is computationally expensive. AKAZE implements fast explicit diffusion (FED) embedded in a pyramidal framework which enhances the speed of feature detection in nonlinear scale space. By using FED schemes, nonlinear scale space can be constructed faster than any other discretization scheme. Furthermore, FED schemes beat AOS (Additive Operator Splitting) schemes in terms of easy implementation and accuracy.

Key points are located by finding the extrema of the second-order derivatives of the image over the nonlinear multi-scale pyramid built from the principle of image diffusion. AKAZE implements M-LDB (Modified-Local

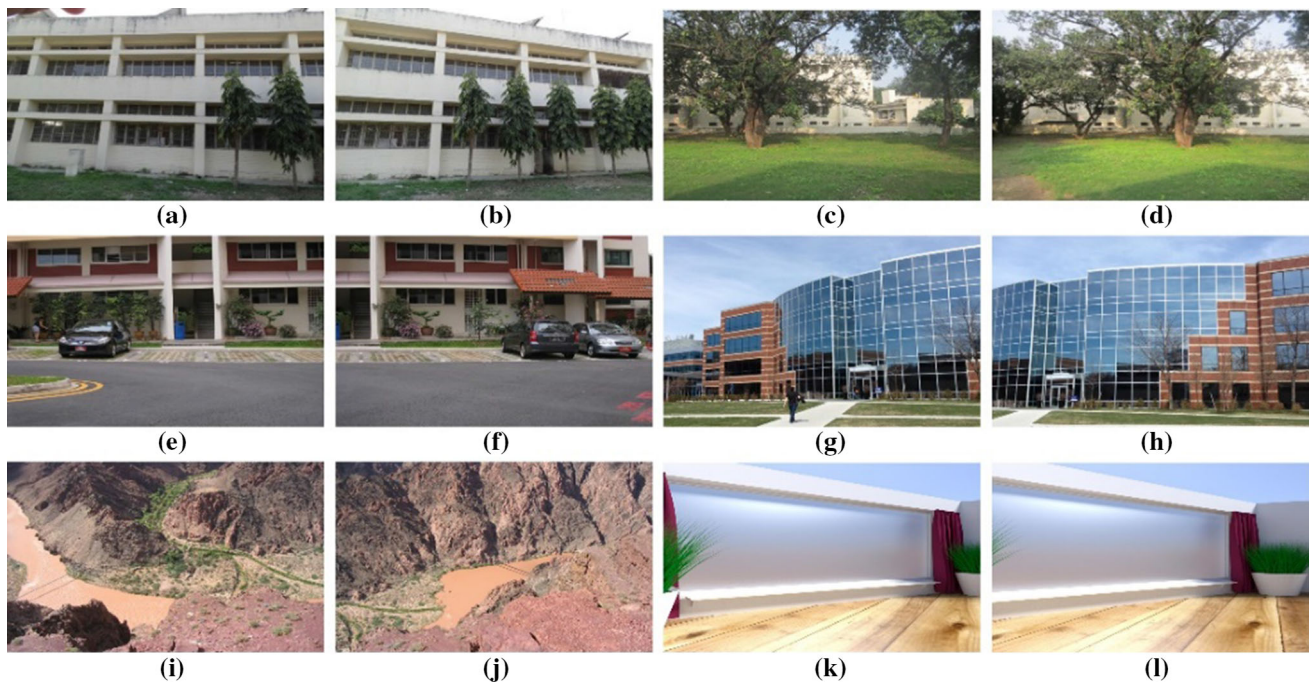


Fig. 1 Input datasets: **a** and **b** building dataset image pair, **c** and **d** tree dataset image pair, **e** and **f** apartment dataset image pair, **g** and **h** MATLAB dataset image pair, **i** and **j** river dataset image pair and **k** and **l** synthetic dataset image pair

Difference Binary) descriptor that is efficient, scale and rotation invariant and requires less storage.

Material and Method

Data Used

In order to test our experiment, we have considered a number of datasets covering different types of scenarios with varying degree of distortions. All the datasets are selected to accomplish different lighting scenarios, feature cover, color type and other constrained imposed due to site (Natural and Manmade)-dependent image capturing conditions. In this study, two datasets have been used: Dataset-A and Dataset-B. Dataset-A is captured by Canon PowerShot handheld digital camera having camera resolution 14 Mega-Pixels. It contains two image pairs: (1) Building dataset and (2) Tree dataset. Dataset-B contains four image pairs downloaded from the internet: (1) Apartment dataset (Zaragoza et al. 2013), (2) MATLAB dataset,¹ (3) Synthetic dataset (Meneghetti et al. 2015) and 4) River dataset.² A subset of these images are shown in Fig. 1.

¹ MATLAB vision toolbox dataset.

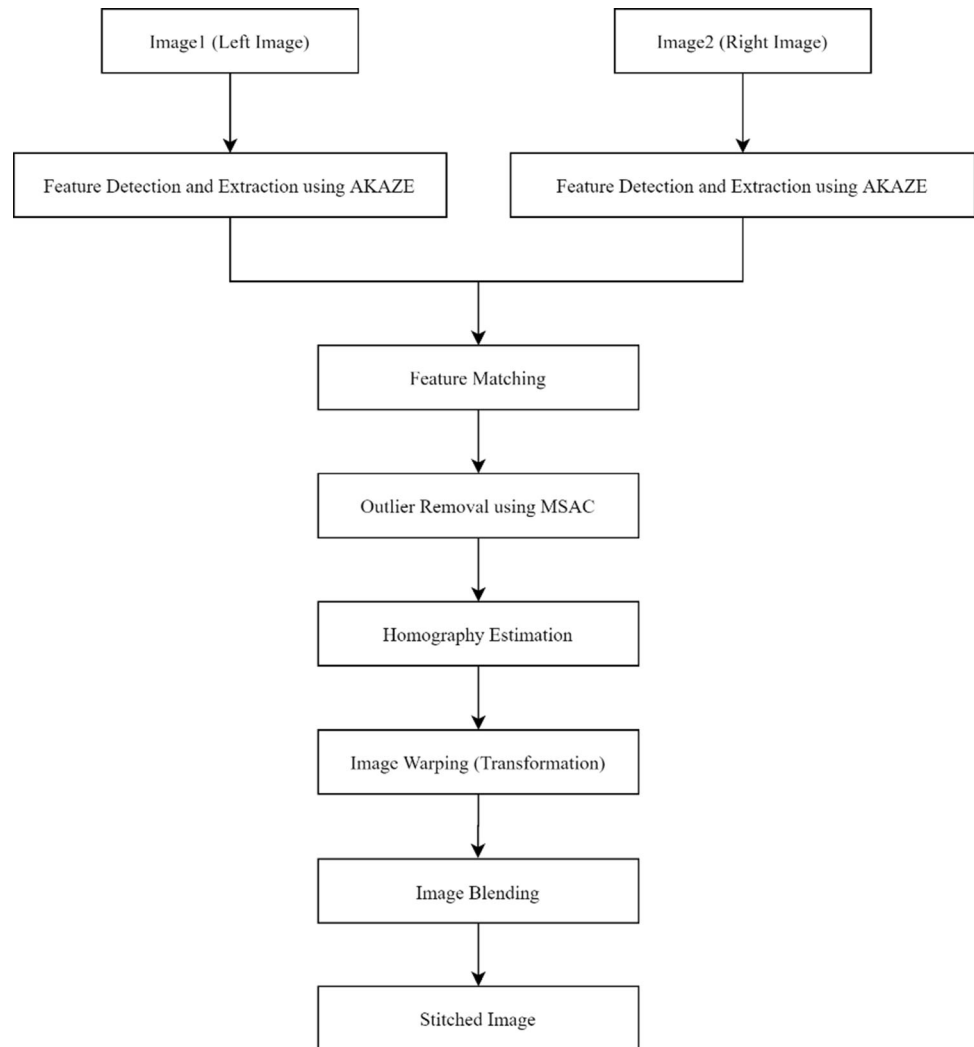
² VLFeat dataset.

Method-Image Stitching Algorithm

The image stitching algorithm proposed is shown in Fig. 2. This algorithm contains mainly two parts: alignment and blending. In alignment part, first, AKAZE is used to detect and extract feature points. Each feature point is matched using k nearest neighbor algorithm which uses k-d tree to find approximate nearest neighbors. Automatic feature matching may give some false matches which can lead to bad alignment, and therefore, false matches must be removed. MSAC is used to remove outliers by minimizing matching error. MSAC is a variant of the Random Sample Consensus (RANSAC) algorithm; Choi et al. (2009) have shown that MSAC is more accurate than RANSAC. The aim of this is to obtain the largest feature points adequate to transformation. The complete process is iterative in nature. The matches obtained by minimizing the matching error are known as the correct matches or inliers. Correct matches are defined as the successful selection of conjugate feature point on the second image for each corresponding feature point on the first image. Next, these inlier points are used to estimate homography (transformation) between two images.

Homography matrix is obtained using the mathematical relation presented in Eq. (1):

$$X' = HX \quad (1)$$

Fig. 2 Image stitching workflow**Fig. 3** Comparison of input and stitched image

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_0 & h_1 & h_2 \\ h_3 & h_4 & h_5 \\ h_6 & h_7 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

where (x, y) and (x', y') are x and y coordinates of corresponding matched points in two images.

After obtaining homography, the target image is warped (transformed) onto the source image. Once images are aligned, there may be visible seam due to different lighting conditions and unavoidable camera movements. Therefore, image blending is used to eliminate the seam. In this paper, we are using the weighted average blending method to eliminate visible seam.

Method-Comparative Analysis

The proposed algorithm is compared with the algorithms which use other detectors such as SIFT, SURF and ORB. Comparison is done based on three parameters: inlier ratio, stitched image quality and execution time.

Inlier Ratio

Inlier ratio is defined as the ratio of the number of correct matches with the number of initial matches.

$$\text{Inlier Ratio} = \frac{\text{\#Correct matches}}{\text{\#Initial matches}}$$

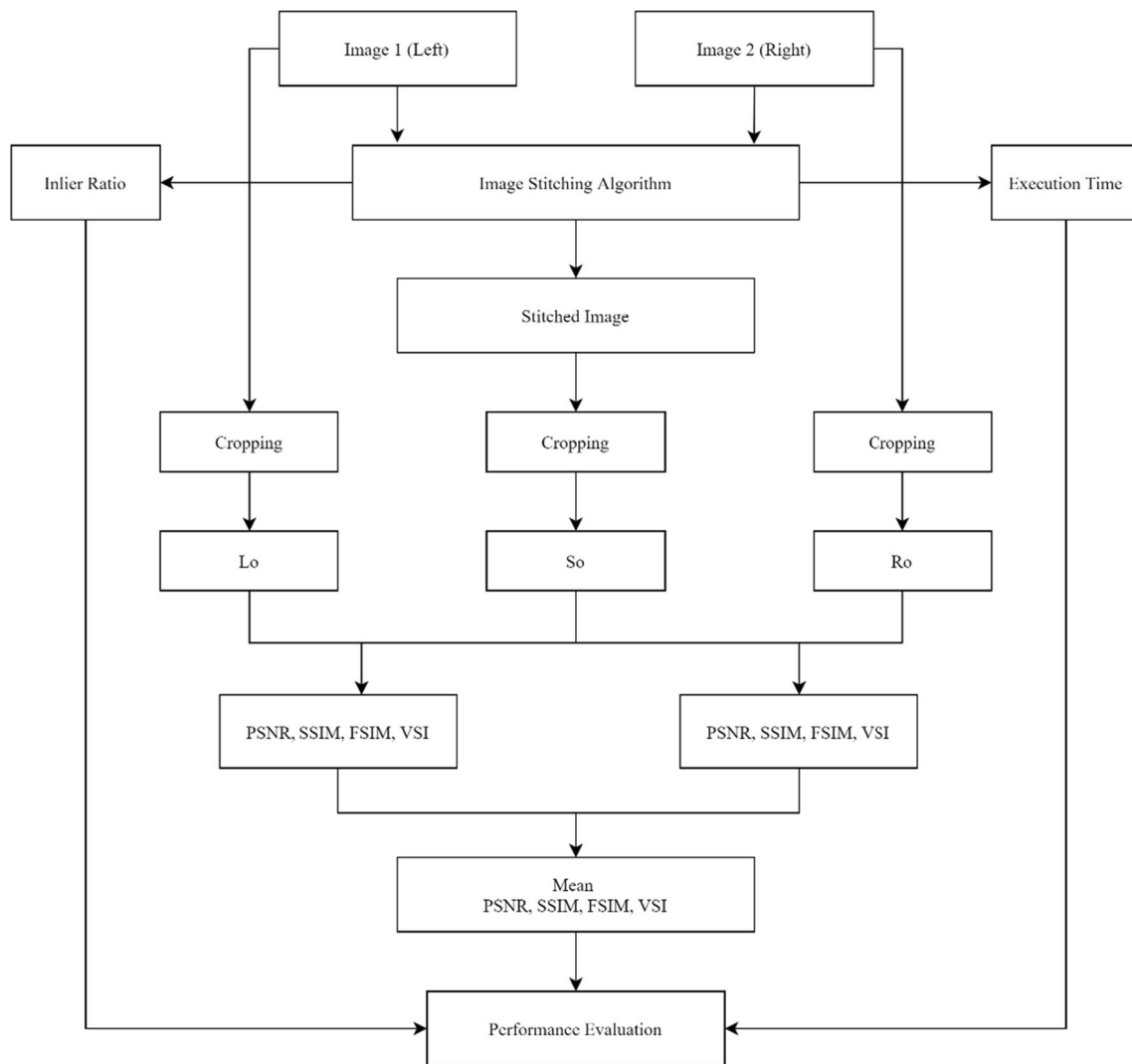


Fig. 4 Workflow for comparison of proposed algorithm with other algorithms

Table 1 Hardware and software specification

Hardware	Processor	Intel(R) Core (TM) i5-7400 CPU @ 3.00GHZ
	Memory	8.0 GB
	Storage (Hard disk)	1 TB
	Operating system	64 bit Operating system
Development resources (Software)	Platform	MATLAB 2017b
	Library	OpenCV-3.1

The number of correct matches affects the quality of final stitched image because homography matrix is calculated using correct matched points which is used in the image alignment step. Therefore, inlier ratio is considered as one of the important parameters to assess the quality of stitched image.

Stitched Image Quality

Mainly, there are two methods followed to evaluate the image quality, the subjective and the objective method. In subjective evaluation method, a number of observers are decided, and then, a number of images are shown to them. Next, they are asked to rate image quality based upon their

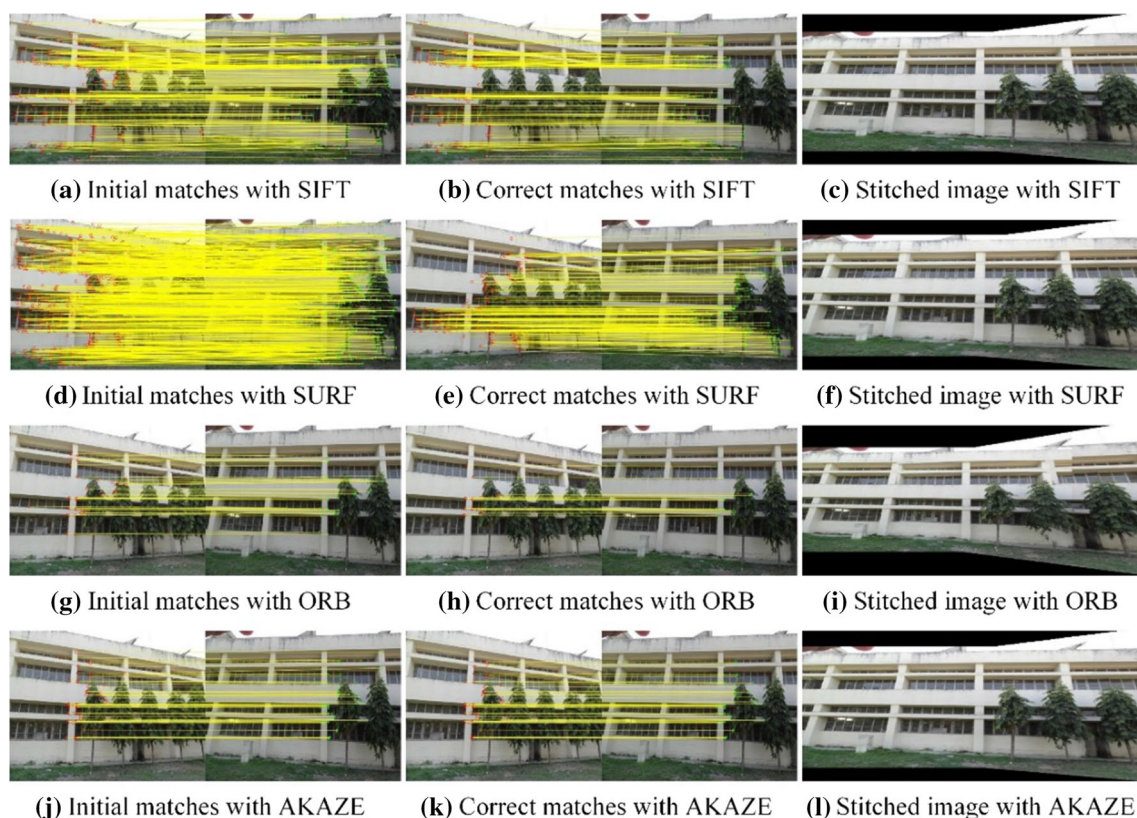


Fig. 5 Building dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

assessment. Each image has been assigned a rank by human inspection based on its quality (Opozda and Sochan 2014). The objective image quality assessment (IQA) methods employ automatic algorithms to evaluate the image quality without human intervention (Opozda and Sochan 2014). As subjective evaluation methods vary person to person and cannot identify minor distortions in the final stitched image. Therefore, in this study we have used objective image quality assessment methods to compare the quality of stitched images. Stitched image obtained using the proposed algorithm is compared with the stitched image obtained using SIFT-, SURF- and ORB-based algorithms. Following are the objective image quality measures used in this work to evaluate the quality of stitched images.

- **Peak Signal-to-Noise Ratio (PSNR)** PSNR is used as a measurement of difference between two images. If the difference between two images is low then PSNR will be high.
- **Structural SIMilarity (SSIM) index** SSIM measures the similarity between two images. Higher the value of SSIM index, lower is the difference between the structure of source and target image (Wang et al. 2004). The image whose quality is being analyzed is called as target image and the image from which the

quality of target image is compared is called as reference image. If there is no structural difference, then $SSIM(\text{source}, \text{target}) = 1$ (Wang et al. 2004).

- **Feature SIMilarity Index (FSIM) index** FSIM index is a full reference IQA method. It is based on the fact that human visual system interprets an image according to its low-level features, higher the FSIM index value better the quality of target image (Zhang et al. 2011).
- **Visual Saliency-Induced Index** Visual saliency (VS) has been widely studied in various areas during the last decade to explore that which areas of an image will draw most attention of the human visual system. Distortions can mostly impact VS maps of images (Zhang et al. 2014). Zhang et. al proposed a simple but very effective full reference image IQA method using VS called visual saliency-induced (VSI) index (Zhang et al. 2014).

The quality of stitched image is evaluated using quality metrics described above. The quality of stitched image generated by each algorithm is evaluated by comparing the quality of overlapping region of input and resultant stitched image. As presented in Fig. 3, for each left image (L) and right image (R) the overlapping areas are compared with the overlapping areas of resultant stitched image. In Fig. 3, Lo , Ro and So represent overlapping region in left image,

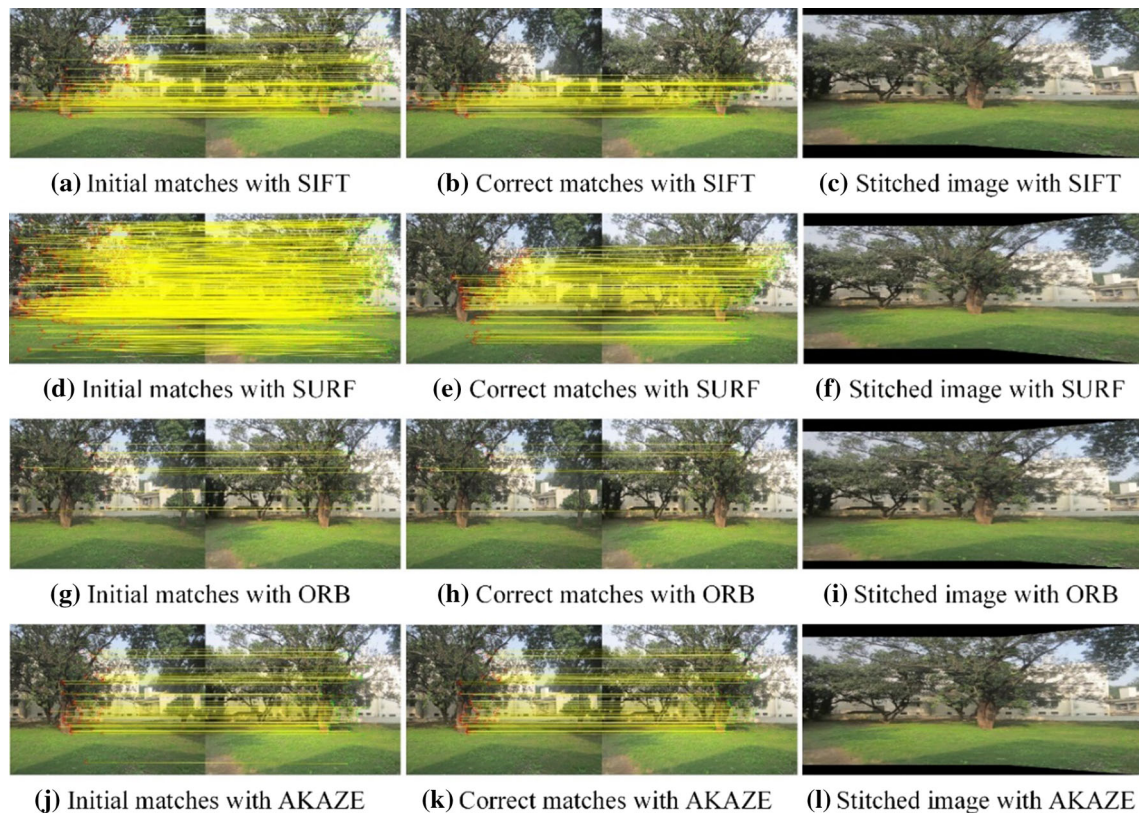


Fig. 6 Tree dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

right image and stitched image, respectively. For each image pair and each algorithm, quality metrics are obtained. For obtaining quality metrics, overlapping region So is compared with Lo and Ro separately and then average value was taken (see Fig. 4). The complete workflow of evaluating the performance of the stitching algorithm is discussed in Fig. 4.

Execution Time

Execution time is the total time taken to detect, match and stitch the input images. Execution time plays important role in selecting image stitching algorithm for real-time applications.

Experimental Environment

Table 1 presents different hardware and software resources used in the current study. The code has been written in MATLAB 2017b using basic image processing and feature detection functions of OpenCV-3.1 library. The development is done on the system having technical specifications provided in Table 1.

Results and Discussion

Figures 5 and 6 show the results obtained using proposed algorithm for building and tree dataset, respectively. The results obtained for apartment dataset and MATLAB dataset are presented in Fig. 7 and 8, respectively. Figure 9 shows the results obtained for river dataset. Finally, the results obtained for synthetic dataset are presented in Fig. 10. Each figure contains 12 sub-figures numbered from (a) to (l). The first row (a–c) in each figure presents the results obtained using SIFT algorithm. Similarly, second, third and fourth rows show the results obtained using SURF, ORB and AKAZE algorithms, respectively. The first column in each row displays initial matched points, and the second column displays correct matched points after removing false matches. The third column shows stitched image obtained using a particular algorithm (SIFT, SURF, ORB, AKAZE).

Stitched images created using various algorithms are shown in the last column of each Figs. 5, 6, 7, 8, 9 and 10. Direct visual assessment of the results defined that the stitched images obtained using AKAZE have better quality than stitched images obtained using other algorithms. The results obtained using SIFT, SURF and ORB contain distortions which are clearly visible when we closely examine

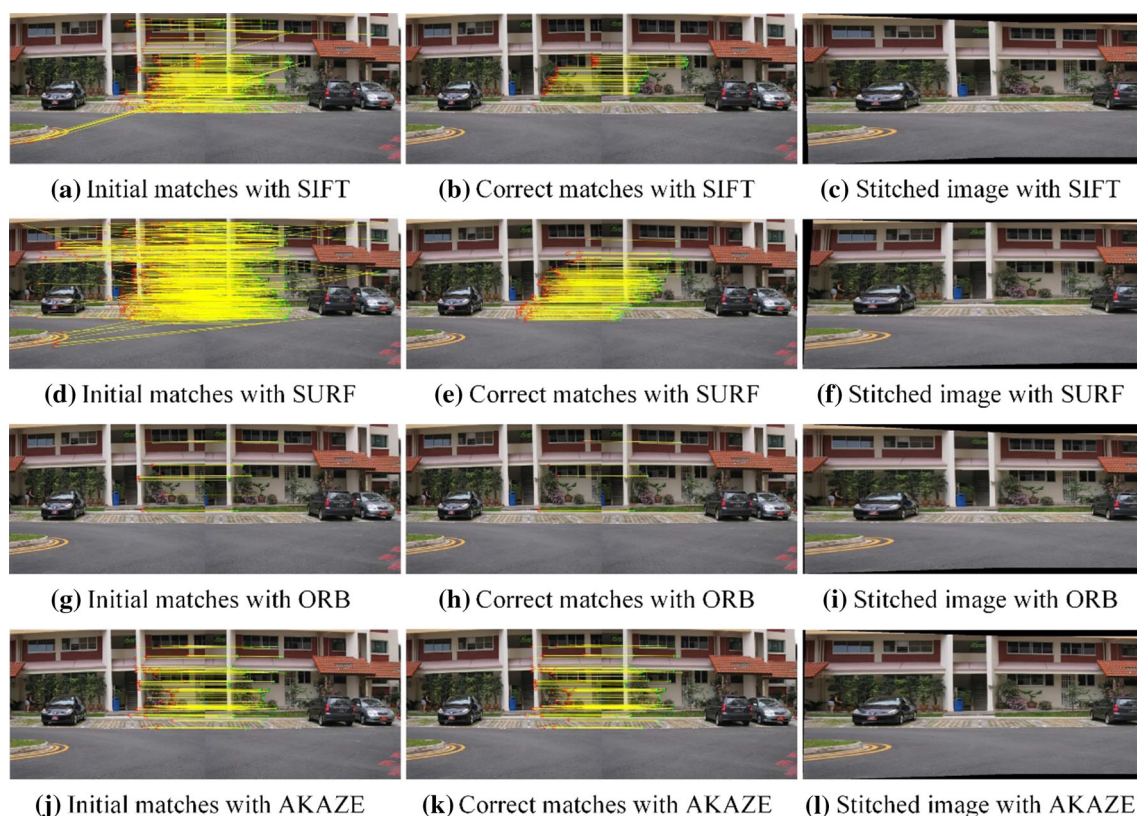


Fig. 7 Apartment dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

some of the outputs. But, in the majority of the cases, visual assessment was not sufficient to describe the minor distortions. Therefore, to precisely evaluate the quality of stitched images and the performance of stitching algorithms, objective image quality assessment methods were used (In “[Stitched Image Quality](#)” section).

Table 2 defines initial and correct matches obtained using SIFT, SURF, ORB and AKAZE algorithms for the selected image datasets. Table 3 presents inlier ratio for each dataset, and also graph for the same is shown in Fig. 11. From Table 3 and Fig. 11, it is observed that AKAZE has the highest inlier ratio in comparison with all other algorithms for all the datasets. For MATLAB dataset, inlier ratio of ORB is slightly higher than AKAZE algorithm. But, a number of initial and correct matches for MATLAB dataset with ORB algorithm are very less (Initial = 12 and Correct = 10). Therefore, in order to get better stitched image quality, inlier ratio should be high and at the same time the number of matches should be more.

From Table 4, building dataset section we can observe that the quality of stitched image obtained using AKAZE-based proposed image stitching algorithm is better than SIFT-, SURF- and ORB-based algorithms for building dataset. For tree dataset also, the quality of stitched image generated using AKAZE-based algorithm is better than all

other algorithms and this can be verified from Table 4, tree dataset section. For river dataset, SURF-based algorithm gives better quality stitched image, and AKAZE-based algorithm comes second. ORB- and SIFT-based algorithms give poor quality in comparison with SURF- and AKAZE-based algorithms. For MATLAB dataset, values of all quality metrics for AKAZE-based algorithm are much higher than all other algorithms which implies that the quality of stitched image obtained using AKAZE-based algorithm is much better than SIFT-, SURF- and ORB-based algorithms. For all other datasets, AKAZE-based algorithm outperforms SIFT-, SURF- and ORB-based algorithms in terms of quality of generated stitched image.

Table 5 and Fig. 12 show the execution time for all datasets using different stitching algorithms. It is observed that ORB-based stitching algorithm is fastest than all other algorithms for all datasets. AKAZE-based algorithm comes second after ORB-based algorithm for all datasets except tree dataset. In tree dataset, SURF overtakes AKAZE-based algorithm in terms of speed. SIFT-based algorithm has better execution time than SURF-based algorithm for all datasets except tree and river dataset. Overall, ORB-based stitched algorithm is fastest, then comes AKAZE, then SIFT and at last SURF-based algorithm.

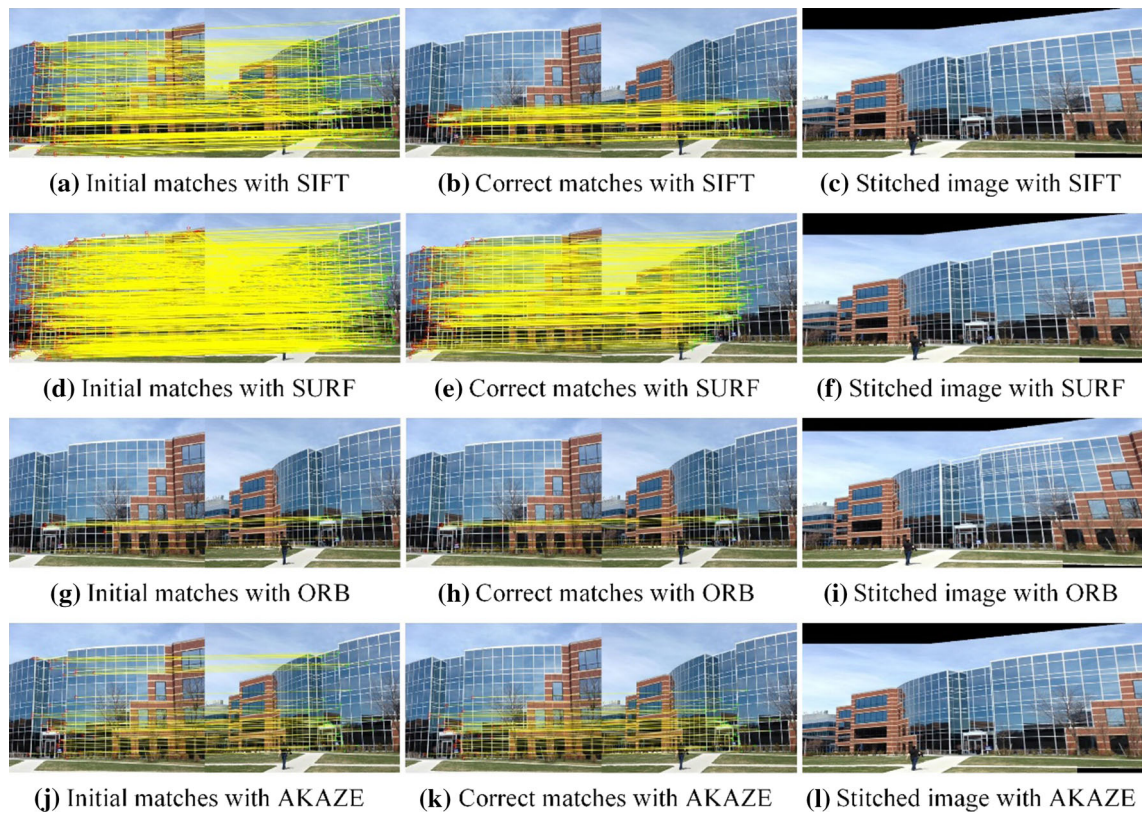


Fig. 8 MATLAB dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

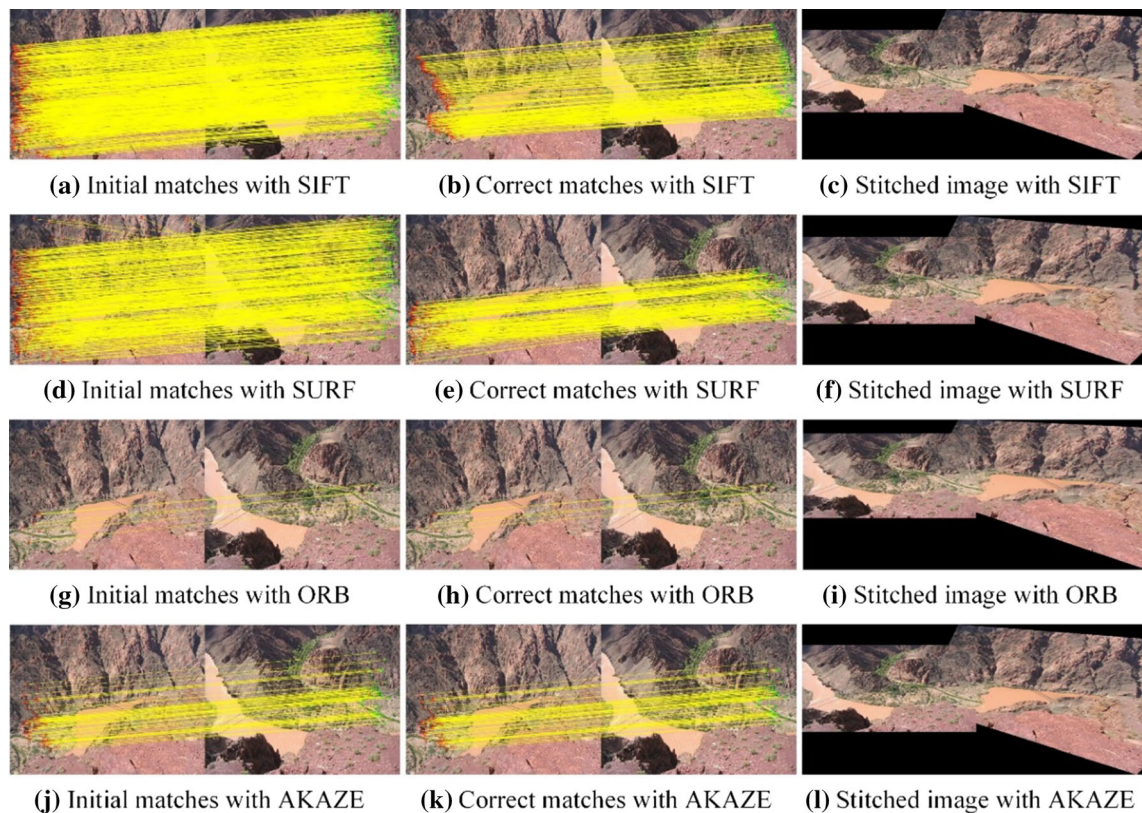


Fig. 9 River dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

Fig. 10 Synthetic dataset: image matching and stitched image using SIFT, SURF, ORB and AKAZE

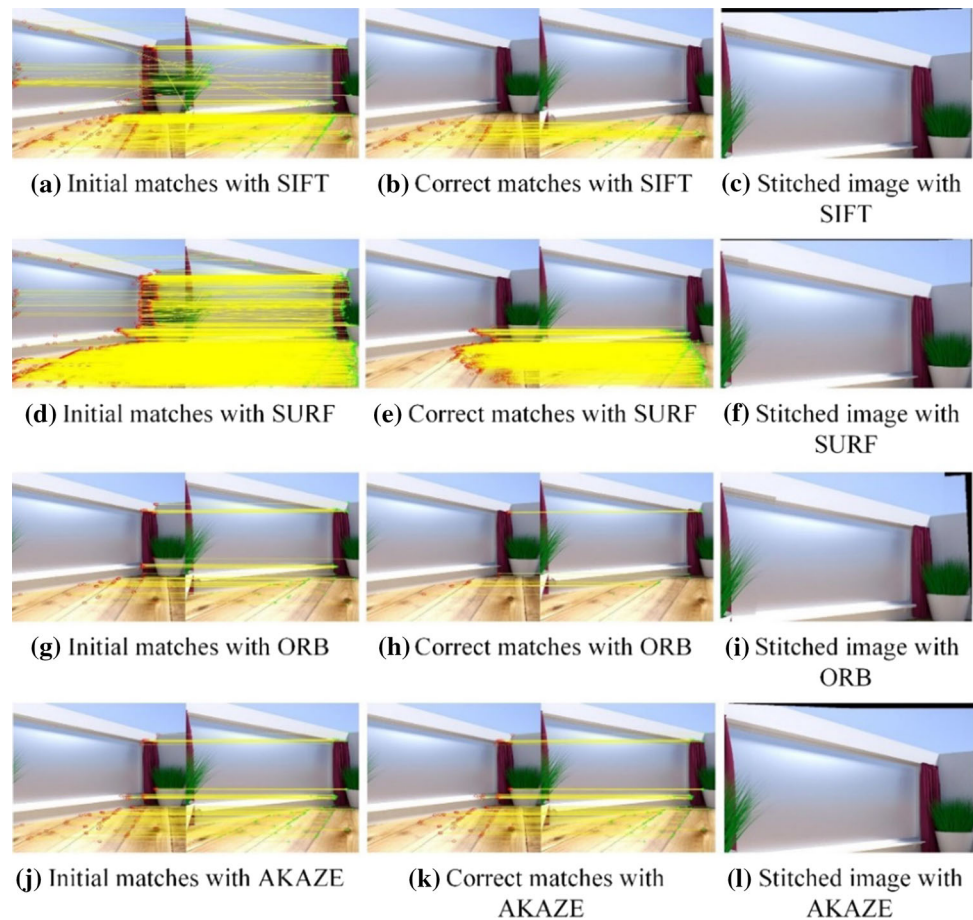


Table 2 Initial and correct matches in selected image dataset

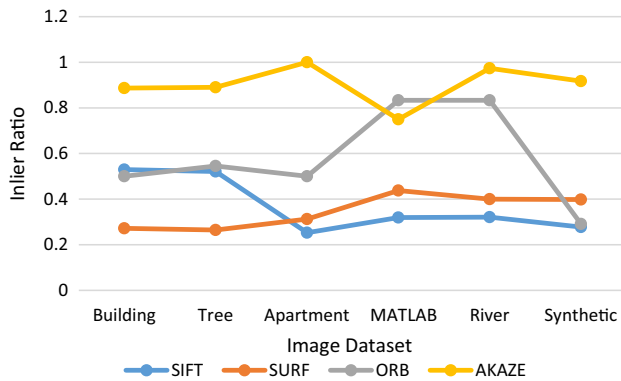
Dataset name	SIFT		SURF		ORB		AKAZE	
	Initial matches	Correct matches	Initial matches	Correct matches	Initial matches	Correct Matches	Initial matches	Correct matches
Building	238	126	740	201	40	20	53	47
Tree	125	64	533	141	11	6	64	57
Apartment	261	66	618	193	16	8	107	107
MATLAB	260	83	768	336	12	10	44	33
River	788	253	493	197	6	5	76	74
Synthetic	281	78	1698	676	86	25	109	100

The current study proposed a new AKAZE detector-descriptor-based stitching algorithm and compared the results with some of the most frequently used image stitching algorithms (SIFT, SURF and ORB). The comparison was done based on three parameters: Inlier ratio, stitched image quality and execution time. It is observed that the inlier ratio for AKAZE-based algorithm is higher than all other algorithms for all image datasets except MATLAB dataset. For MATLAB dataset, AKAZE has the second best inlier ratio after ORB but ORB has very less

number of matched points which affects stitched image quality. In terms of quality of stitched image, AKAZE-based algorithm outperforms all other stitching algorithms for all the datasets except for river dataset. For river dataset, quality of stitched image generated using SURF-based algorithm is better than quality of stitched image generated using AKAZE-based algorithm. It is also observed that execution time of ORB-based algorithm is fastest for all the datasets but the quality of generated stitched images is very poor. AKAZE-based algorithm

Table 3 Inlier ratio in selected image dataset

Dataset name	SIFT	SURF	ORB	AKAZE
Building	0.5294	0.2716	0.5	0.8868
Tree	0.521	0.2645	0.5455	0.8906
Apartment	0.2529	0.3123	0.5	1
MATLAB	0.3192	0.4375	0.8333	0.75
River	0.3211	0.3996	0.8333	0.9737
Synthetic	0.2776	0.3981	0.2907	0.9174

**Fig. 11** Comparative assessment of Inlier ratio in selected image dataset

comes second in terms of faster execution time for all datasets which defines its computational efficiency.

Overall, it is observed that the proposed AKAZE-based algorithm shows better image stitching performance in comparison with all the other algorithms. SIFT, SURF, ORB algorithms identify and define features at different scale intervals by developing or approximating the Gaussian scale space of an individual image. But, Gaussian blurring does not recognize the natural edges of objects and smoothes to an identical degree, both features and noise, diminishing localization accuracy and distinctiveness. Due to which they contain more false matches, that in turn reduces inlier ratio. In distinction, AKAZE recognizes and defines 2D features in a nonlinear scale space with nonlinear diffusion filtering. In this manner, AKAZE can cause blurring regionally adaptive to the image dataset, it diminishes noise but preserves object boundaries, achieving superior localization accuracy and distinctiveness. Because of better localization accuracy and distinctiveness in feature detection and description, AKAZE produces less number of false matches which increases its inlier ratio and better image stitching quality.

Table 4 Stitched image quality comparison

<i>Building dataset</i>				
	PSNR	SSIM	FSIM	VSI
SIFT	15.290	0.50829	0.71583	0.91124
SURF	15.730	0.50982	0.71323	0.91253
ORB	13.967	0.44804	0.67129	0.90118
AKAZE	15.474	0.57036	0.75763	0.92592
<i>Tree dataset</i>				
SIFT	15.365	0.48183	0.67456	0.89163
SURF	15.372	0.47199	0.67295	0.89020
ORB	15.199	0.46817	0.67186	0.88135
AKAZE	15.402	0.48256	0.67790	0.89978
<i>Apartment dataset</i>				
SIFT	15.925	0.50463	0.66008	0.87245
SURF	15.631	0.50009	0.64441	0.86540
ORB	15.403	0.49780	0.64184	0.86404
AKAZE	15.758	0.51946	0.67334	0.87866
<i>MATLAB dataset</i>				
SIFT	18.445	0.80445	0.82259	0.94158
SURF	20.089	0.87435	0.75784	0.95644
ORB	15.210	0.64714	0.64348	0.90531
AKAZE	21.488	0.91599	0.88792	0.97491
<i>River dataset</i>				
SIFT	16.382	0.34612	0.71573	0.88504
SURF	17.203	0.35358	0.72362	0.89361
ORB	16.257	0.29659	0.70566	0.88219
AKAZE	16.814	0.35285	0.71811	0.88900
<i>Synthetic dataset</i>				
SIFT	25.103	0.89256	0.87040	0.95703
SURF	24.230	0.87156	0.84548	0.94123
ORB	22.861	0.85353	0.83096	0.93538
AKAZE	25.718	0.89466	0.87272	0.96433

Table 5 Execution time for all datasets

Dataset name	SIFT	SURF	ORB	AKAZE
Building	2.30314	2.34363	1.97112	2.00927
Tree	3.00375	2.52199	1.89067	2.81753
Apartment	2.10060	2.18485	1.92716	1.97592
MATLAB	1.98930	2.39048	1.70119	1.86169
River	5.65870	5.31550	3.06651	3.61165
Synthetic	4.91838	5.01601	3.94602	4.84406

Conclusions

The paper has presented an image stitching algorithm based on AKAZE-based features. In image stitching algorithm, we used AKAZE for feature detection and K-NN for

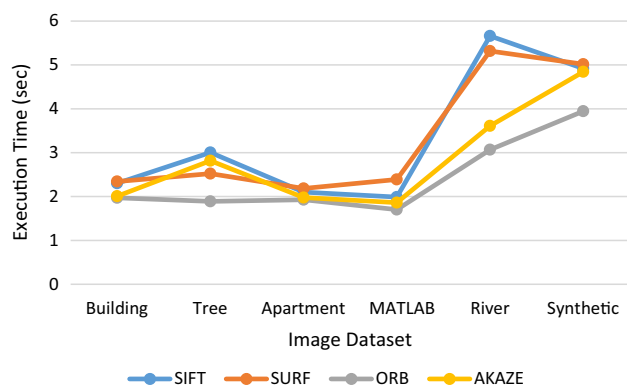


Fig. 12 Comparative assessment of execution time in selected image dataset

matching detected points in overlapping images. MSAC algorithm is used to remove false matches, and finally weighted average blending is used to blend matched images. We also compared our AKAZE-based image stitching algorithm with other popular algorithms, namely SIFT-, SURF- and ORB-based image stitching algorithms on different types of datasets. In our experimental results, we found that ORB-based image stitching implementation is faster than other methods but it degrades the quality of final stitched image. SIFT-based implementation provides good quality of stitched image, but it is computationally slower. The newly proposed AKAZE-based stitching algorithm provides good quality of stitched image in comparison with all other algorithms, and also it is computationally faster than SIFT- and SURF-based stitching algorithms.

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